SURFACE ENGINEERING
Wear Resistance Specialists

Cobalt #21 Data Sheet

Description: Cobalt alloy #21 is a Molybdenum strengthened, high temperature, corrosion resistant alloy with exceptional impact, erosion , and metal to metal wear resistance. The alloy work hardens in service. It has moderate abrasion resistance as compared to Seaco Cobalt alloy #6.

Applications: Hot Forming Dies, Forging Dies, Erosion Shields, Valve Seats and plugs, Hydro-Turbine Runners, Pump shafts and sleeves.

Typical Chemical Analysis:

C	Mn	Si	Ni	Cr	W	Mo	Fe	Co
.25	.50	.60	2.5	26	-	5.5	1.0	Bal.

Machineability: Good

Hardness: Rc 22-32 as deposited, Rc 38-40 wok hardened

Hot Hardness (DPH): 800°F - 130; 1000°F - 135; 1200°F - 140; 1400°F - 110

Adhesive Wear Data (Avg.): 90lb Load - 2.5mm³; 150lb Load - 5.2mm³
(Volume Loss) 210lb Load - 10.3mm³; 300lb Load - 14.5mm³

Abrasive Wear Data: Volume Loss of 86.4mm³ on ASTM G65B Test

Charpy Impact Strength (Un-Notched, Room Temperature): 27 Ft. Lb

Density: .299 Lbs/In³

Liquidus: 2480°F